Nozzle Heat Transfer Model – For Theseus LPL Engine

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1. Introduction

One of the most critical parts of a rocket engine is the nozzle, and needs to withstand the heat transfer, so the inside nozzle wall should stay lower than the material melting point.

In this document, a detailed procedure of how to model heat transfer focusing on the Theseus nozzle (is a heat sink nozzle) will be presented. All plots are from the analysis of the current Thesus copper-alloy nozzle.

2. Getting the necessary input data

2.1. Mach number

First, the Mach number along the nozzle is calculated from the geometry input.

$$\varepsilon_{nozzle} = \frac{1}{M}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{2.1}$$

where ε_{nozzle} is the expansion ratio for each position.

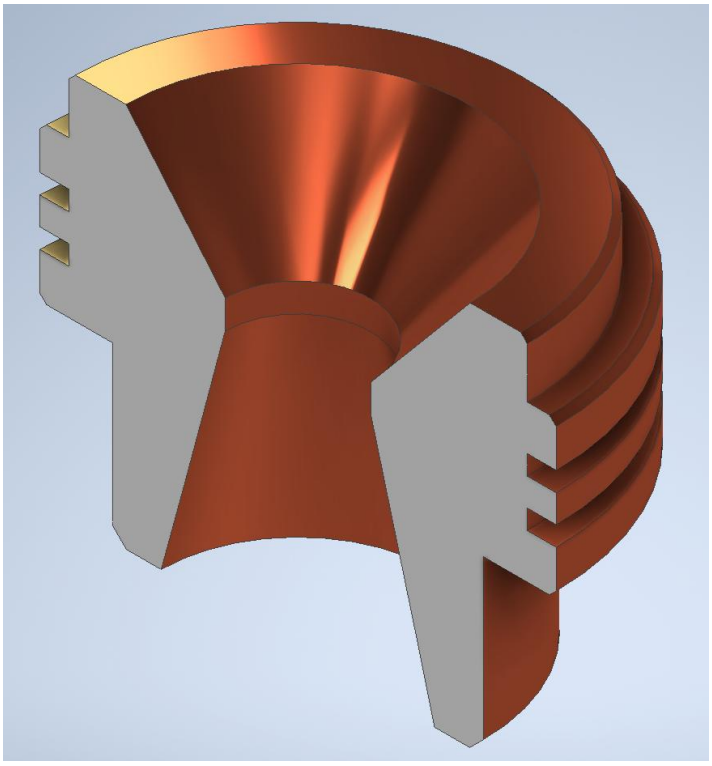

Figure 2.1: Theseus copper-alloy nozzle.

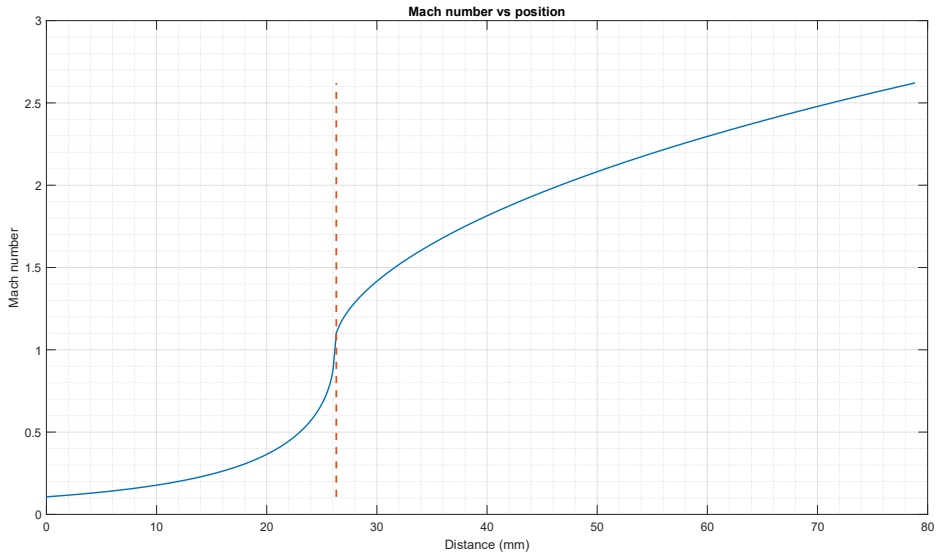

Figure 2.2: Mach number vs position.

Dashed line means throat zone.

2.2. Recovery wall temperature

Recovery wall temperature, also known as adiabatic wall temperature, is the temperature a surface would reach if there was no heat transfer or radiation exchange. It considers the temperature drop produced by the heat boundary layer in the wall, i.e., the temperature that the wall sees. It follows this expression:

$$T_r = T_\infty \left(1 + r\frac{\gamma - 1}{2}M_\infty^2\right) \tag{2.2}$$

where ∞ means conditions at the free stream zone.

r is the recovery factor, and for turbulence flow, as is the case, $r \propto \mathrm{Pr}^{1/3}$. For a preliminary analysis we will consider that:

$$r = Pr^{1/3} \tag{2.3}$$

where Pr is the Prandtl number.

As a first approximation Prandtl number can be considered constant from frozen values, so we will take its value from CEA calculations at the nozzle throat, i.e., $\boldsymbol{Pr = 0.5473}$.

Equating (2.2) with (2.3):

$$T_r = T_\infty \left(1 + Pr^{1/3}\frac{\gamma - 1}{2}M_\infty^2\right) \tag{2.4}$$

This value is found from CEA output in the following section of the output file:

```
 WITH FROZEN REACTIONS

Cp, KJ/(KG)(K)     2.2121   2.1812   2.0169   2.1419   2.1050   2.0789   2.0583   2.0413
CONDUCTIVITY       3.4040   3.0902   2.0171   2.7596   2.5006   2.3404   2.2251   2.1363
PRANDTL NUMBER     0.5439   0.5473   0.5455   0.5497   0.5504   0.5499   0.5490   0.5479
```

Figure 2.3: CEA output Prandtl number.

where the value in the red box is the Prandtl number of the throat, which as first approximation will be considered constant across the nozzle as it can be seen.

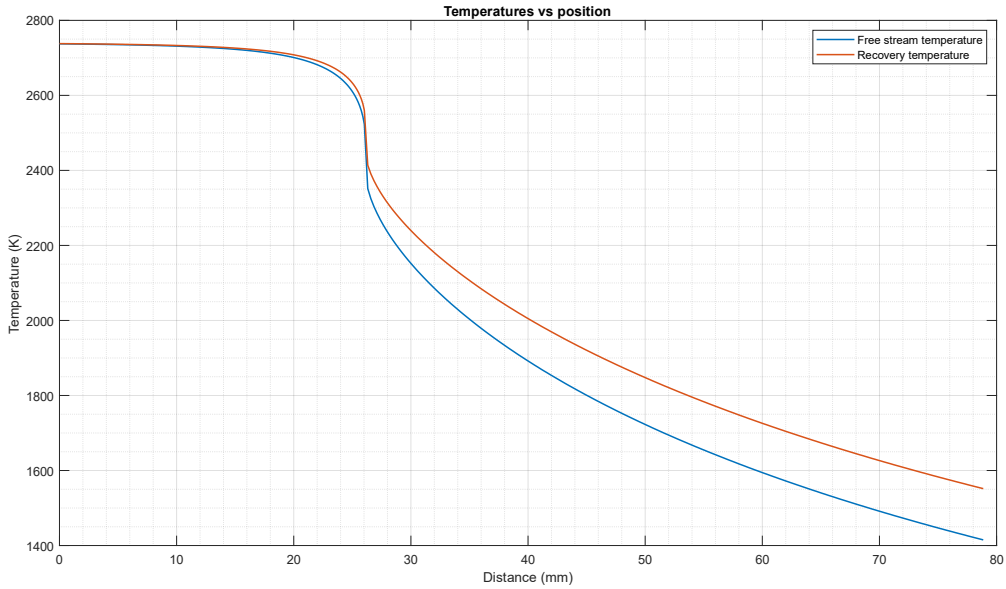

Figure 2.4: Recovery temperature.

2.3. Convective heat transfer coefficient

Convective heat transfer (h_g) is obtained from Nusselt number (Nu). This Nusselt number is a general expression used in nozzle, it is the general expression of the Bartz correlation nozzle heat transfer.

$$Nu = 0.026 Re^{0.8} \mathrm{Pr}^{0.4} \left(\frac{\rho_{am}}{\rho_\infty}\right)\left(\frac{\mu_{am}}{\mu_0}\right) \tag{2.5}$$

$$h_g = \frac{Nu \cdot k}{D} \tag{2.6}$$

where D is the diameter at each section of the divergence nozzle area.

Subscript am mean arithmetic means conditions, so:

$$\rho_{am} = \frac{1}{2}[\rho_{wall} + \rho_\infty] \tag{2.7}$$

$$\mu_{am} = \frac{1}{2}[\mu_{wall} + \mu_\infty] \tag{2.8}$$

Also,

$$\mu_0 = \mu(T_0) \tag{2.9}$$

where T_0 is the stagnation temperature, which is considered constant in an isentropic nozzle expansion, so μ_0 is the viscosity at the chamber with the adiabatic temperature T_{ad}.

Viscosity variation is extrapolated from a few points, so extrapolation follows the following equation:

$$\mu = \mu_{ref}\left(\frac{T}{T_{ref}}\right)^{\omega} \tag{2.10}$$

where ω is obtained from computing logarithmic interpolation, and we consider reference as the properties from the throat.

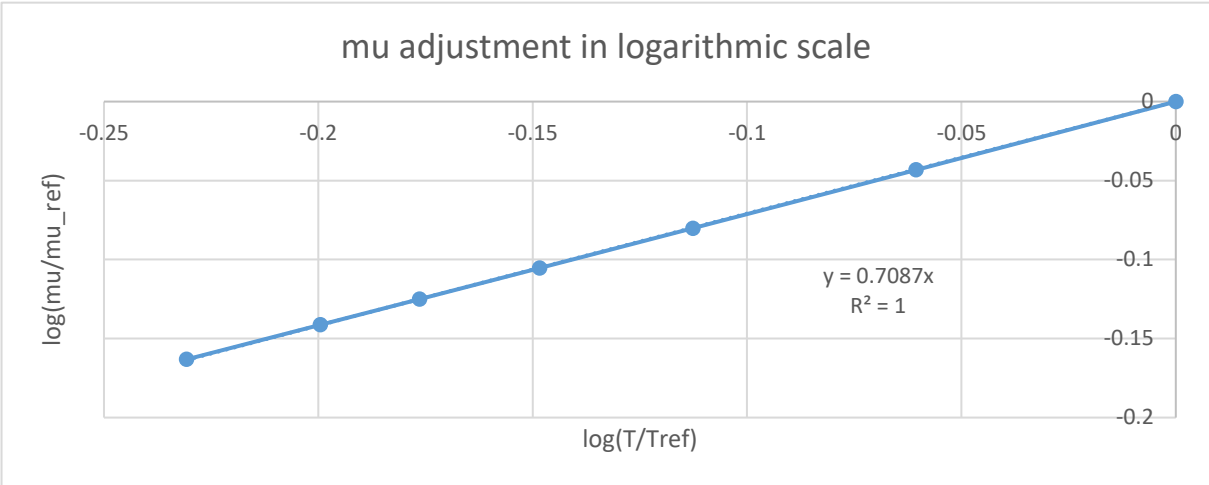

Figure 2.5: μ linear fitting in logarithmic scale.

In this case, we can conclude that $\omega = 0.7087$.

We need to calculate Reynold number at each divergent nozzle section, so:

$$v = M\sqrt{\gamma R_g T} \tag{2.11}$$

$$P = P_{ch}\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-\frac{\gamma}{\gamma-1}} \tag{2.12}$$

$$\rho = \frac{P}{R_g T} \tag{2.13}$$

$$Re = \frac{\rho v D}{\mu} \tag{2.14}$$

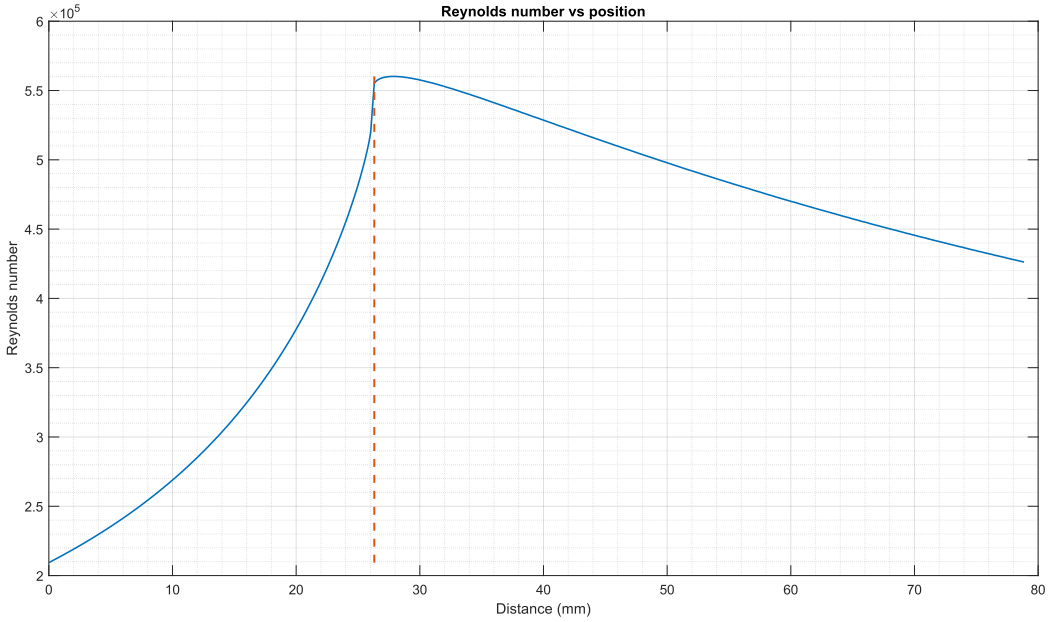

Figure 2.6: Reynolds number along divergent nozzle section.

As Reynolds' number is decreasing, boundary layer thickness will increase along the divergent nozzle section.

For calculating the fluid density at the wall, we need to use the wall temperature from Equation (2.4), so:

$$\rho_{wall} = \frac{P_{wall}}{R_g T_{wall}} \tag{2.15}$$

where $P_{wall} = P$, and $T_{wall} = T_r$.

The pressure distribution throughout the boundary layer in the direction normal to the surface remains relatively constant throughout the boundary layer and is the same as on the surface itself. This is why $P_{wall} = P$.

μ_{wall} is calculated by using Equation (2.10), but now the temperature used is the recovery temperature from Equation (2.4).

So now we can calculate mean conditions for density and viscosity using Equations (2.7) and (2.8).

Using the constant Prandtl number assumption value from frozen conditions at throat, i.e.:

$$Pr = \frac{\mu C_p}{k} = \frac{\mu_{ref} C_{p_{ref}}}{k_{ref}} \approx constant \tag{2.16}$$

In Equation (2.16) we have two unknowns, i.e., μ and Cp, so as a first approximation this problem can be solved saying that $C_p \approx Cp_0$, so:

$$k = k_{ref} \frac{\mu}{\mu_{ref}} \tag{2.17}$$

where k_{ref} is the thermal conductivity at throat taken as a reference (throat), and μ_{ref} is the viscosity at throat taken as a reference.

k_{ref} is taken from CEA section; see orange box from Figure 2.3. Bear in mind that the units are in $\frac{mW}{cm \cdot K}$, so you should divide by 10 to convert to $\frac{W}{m \cdot K}$.

μ_{ref} can be taken from the following CEA section:

```
TRANSPORT PROPERTIES (GASES ONLY)
  CONDUCTIVITY IN UNITS OF MILLIWATTS/(CM)(K)

VISC,MILLIPOISE  0.83705  0.77544  0.54553  0.70827  0.65381  0.61910  0.59352  0.57337
```

Figure 2.7: Viscosity.

Nusselt number and conductivity along the nozzle looks as follows:

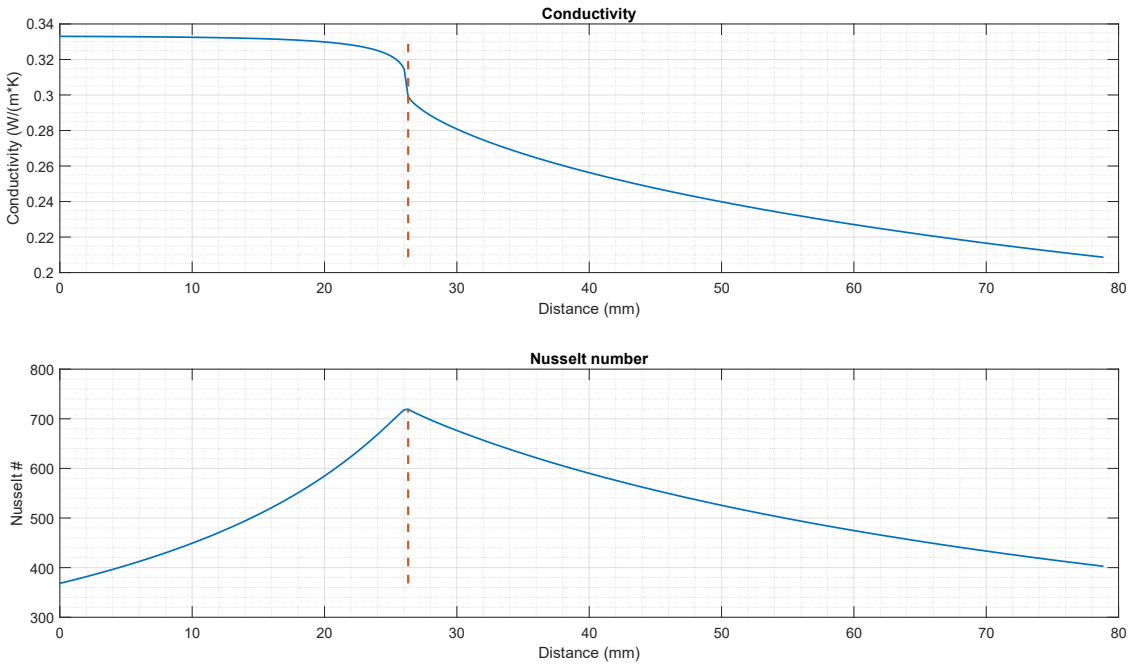

Figure 2.8: Nusselt number and conductivity.

Finally, by using Equation (2.6) the heat transfer coefficient looks as follows:

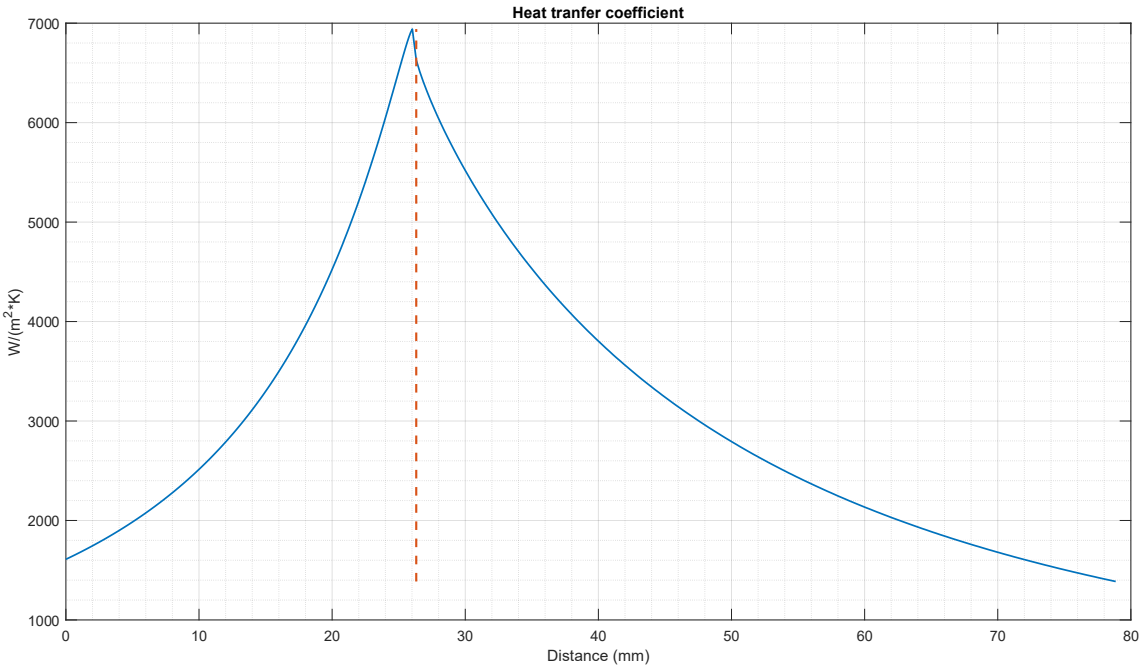

Figure 2.9: Heat transfer coefficient along divergent nozzle section.

2.4. Summary of the input data needed

For more accuracy, an isentropic coefficient when Mach number is less than one is chosen (the one in the combustion chamber), and another one when Mach number is great than one (average from the divergent section)

In the following tables, a summary of the data needed to simulate the heat transfer in the Theseus engine will be shown.

Bear in mind that geometry nozzle input will be needed to compute the Mach number.

Also, variables from the next chapters will be shown, i.e., all the variables needed to compute the heat transfer in the nozzle.

Table 2.1: Thermodynamic properties.

Variable [units]	Physical name	Values for Theseus engine
R_g [J/(Kg*K)]	Specific gas constant	448.436
Pr [-]	Prandtl number	0.5473
μ_0 [mP]	Chamber viscosity	0.83705
ω [-]	Viscosity adjustment parameter	0.7087
μ_{ref} [mP]	Reference viscosity (throat)	0.77544

T_{ref} [K]	Reference temperature (throat)	2463.22
k_{ref} [W/(m*K)]	Reference conductivity (throat)	0.30902
T_{ch} [K]	Chamber temperature	2740.87
P_{ch} [Bar]	Chamber pressure	21.6978
γ_{ch} [-]	Chamber isentropic coefficient	1.2226
γ_d [-]	Average divergent isentropic coefficient	1.2727

Table 2.2: Nozzle B95-CuCo2Be material properties.

Variable [units]	Physical name	Values for Theseus engine
ρ [Kg/m^3]	Nozzle density	8850
c [J/(kg*K)]	Nozzle specific heat	420
k [W/(m*K)]	Nozzle thermal conductivity	195
ε [-]	Nozzle surface emissivity	0.85

3. Model assumptions

- Ideal gas and isentropic nozzle.
- Heat transfer for the longitudinal direction can be neglected; this makes the model to be more conservative and easier to model.
- Chemical equilibrium is achieved from convergent to throat area, and frozen properties from throat to exit area. (CEA from NASA is used for variable fluid properties across the nozzle).
- Nozzle thermal diffusivity is considered constant and worst values for security margin are considered.
- Nusselt number from Bartz correlation for nozzle is used.
- Recovery factor exponent of 1/3 is used in the boundary layer, as it is a turbulence boundary layer.
- Gas that goes through the nozzle does not cool down due to heat exchange with wall because of the short residence time.
- No heat transfer from the nozzle to the ambient in order to be conservative.

4. Heat fluxes

The heat transfer comes from heat fluxes. There are two main fluxes that go into the walls. One is the convective heat flux (q_{con}), and the other is the radiation heat flux (q_{rad}). They have the following expressions and curve shapes.

$$q_{con} = h_g(T_{ch} - T_s) \tag{4.1}$$

$$q_{rad} = \sigma\varepsilon\left(T_{ch}^4 - T_s^4\right) \tag{4.2}$$

where σ is the Stefan-Boltzmann constant, and ε is the surface emissivity, and T_s is the nozzle inside surface temperature.

So, the total heat flux that goes into the nozzle wall is:

$$q_{tot} = q_{con} + q_{rad} \tag{4.3}$$

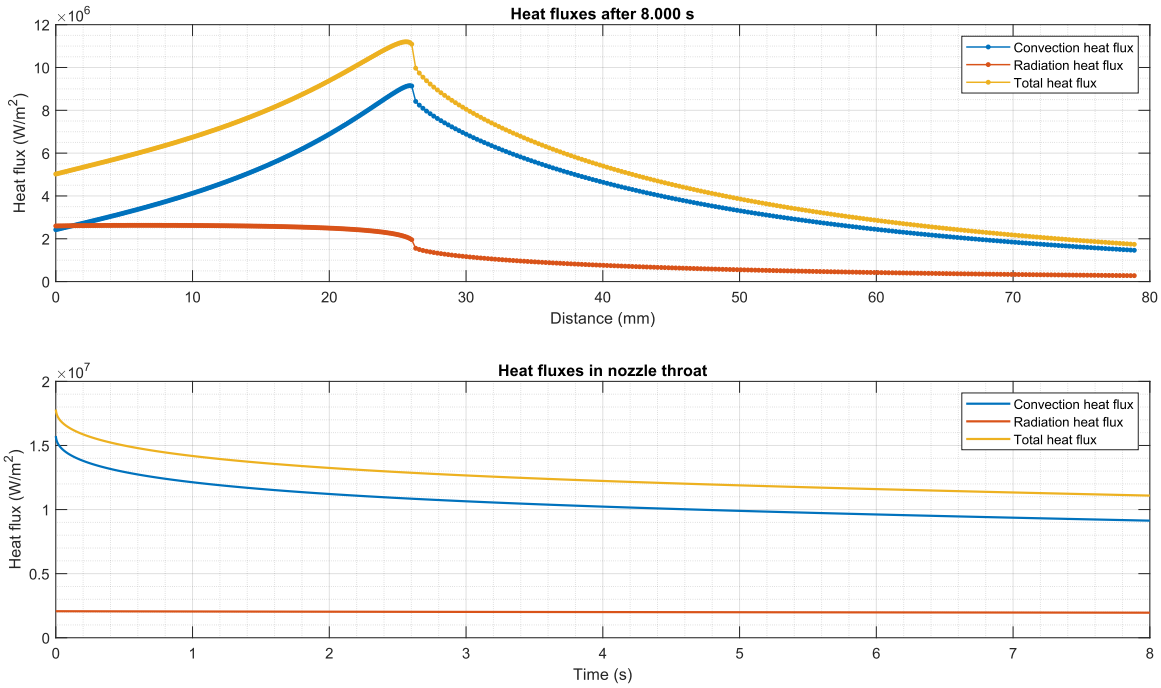

Figure 4.1: Heat flux nozzle wall obtained after calculations.

5. Conduction in the nozzle

Convection and radiation are solved with equations (4.1) and (4.2), respectively. Now it is time to solve the conduction through the nozzle material and complete the problem, since solving the heat flux also requires knowing the nozzle wall temperature, and vice versa.

For this, the conduction equation in polar coordinates is used, with the following boundary and initial conditions.

$$\frac{\partial T}{\partial t} = \frac{\alpha}{r}\frac{\partial T}{\partial r} + \alpha\frac{\partial^2 T}{\partial r^2} \tag{5.1}$$

$$k\frac{\partial T}{\partial r}\bigg|_{r=R_{int}} = -q_{tot} \tag{5.2}$$

$$\frac{\partial T}{\partial r}\bigg|_{r=R_{ext}} = 0 \tag{5.3}$$

where $\alpha = \frac{k}{\rho c}$ is the thermal diffusivity.

This equation needs to be solved numerically, so discretization is needed.

The domain where the equation will be solved goes from the spatial point $j = 1$ to $j = J$. Also, for each transversal slide (different R_{int} and R_{ext}) a different problem with different diameters needs to be solved. The distance between each j is Δr.

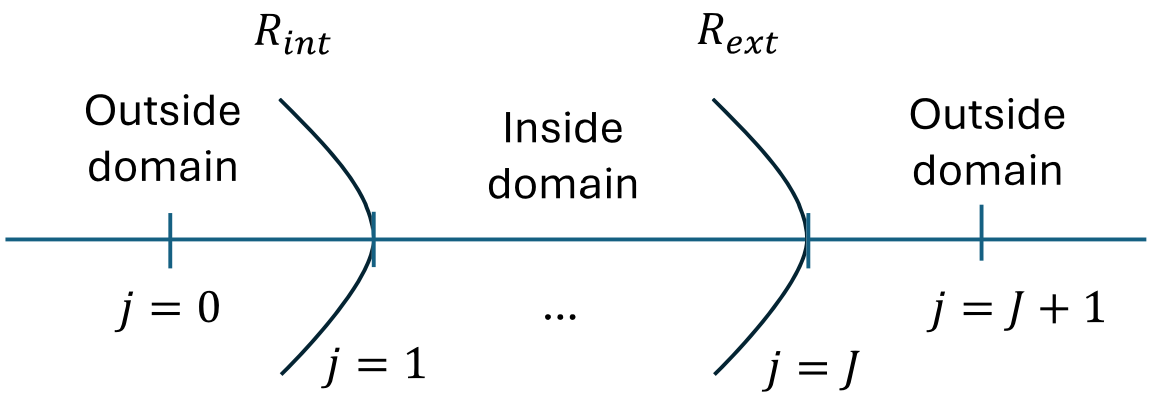

Figure 5.1: Scheme of the discretized domain.

The equation and the boundary conditions are discretized with an explicit time and a centered space, that looks as follows.

$$\frac{T_j^{n+1} - T_j^n}{\Delta t} = \frac{\alpha}{r_j}\frac{T_{j+1}^n - T_{j-1}^n}{2\Delta r} + \alpha\frac{T_{j+1}^n - 2T_j^n + T_{j-1}^n}{\Delta r^2} \tag{5.4}$$

$$k\frac{T_2^n - T_0^n}{2\Delta r} = -h_g(T_{ch}^n - T_1^n) - \sigma\varepsilon[(T_{ch}^n)^4 - (T_1^n)^4] \tag{5.5}$$

$$\frac{T_{J+1}^n - T_{J-1}^n}{2\Delta r} = 0 \tag{5.6}$$

where Δt is the time increment, and n is each time element.

If j is inside the domain, then, from Equation (5.4):

$$T_j^{n+1} = sa_j T_{j-1}^n + (1-2s)T_j^n + sb_j T_{j+1}^n \tag{5.7}$$

where

$$s \doteq \frac{\alpha \Delta t}{\Delta r^2} \tag{5.8}$$

$$a_j = 1 - \frac{\Delta r}{2r_j} \tag{5.9}$$

$$b_j = 1 + \frac{\Delta r}{2r_j} \tag{5.10}$$

s is a particularly important parameter for numerical stability. For converge to a solution is especially important to ensure that $s \leq \frac{1}{2}$.

Also, bear in mind that if $s = \frac{1}{6}$, then the numerical error of this numerical scheme is $\varepsilon_j^n \sim o(\Delta t^2, \Delta r^2)$, else $\varepsilon_j^n \sim o(\Delta t, \Delta r^2)$.

If $j = 1$, i.e., the inside surface of the nozzle, then:

$$T_1^{n+1} = sa_j T_0^n + (1-2s)T_1^n + sb_j T_2^n \tag{5.11}$$

Using Equation (5.5) for T_0^n, then

$$T_1^{n+1} = 2sT_2^n + (1-2s)T_1^n + sa_j \frac{2\Delta r h_g}{k}(T_{ch}^n - T_1^n) + sa_j \frac{2\Delta r \sigma \varepsilon}{k}[(T_{ch}^n)^4 - (T_1^n)^4] \tag{5.12}$$

If $j = J$, i.e., the outside surface of the nozzle, then, by equating (5.4) and (5.6):

$$T_J^{n+1} = 2sT_{J-1}^n + (1-2s)T_J^n \tag{5.13}$$

For initializing the calculations, the initial condition of temperature in the nozzle, equal the atmospheric temperature, is implemented.

6. Results

Computing all the previous equations for the values in Table 2.1 and Table 2.2, the following results are obtained.

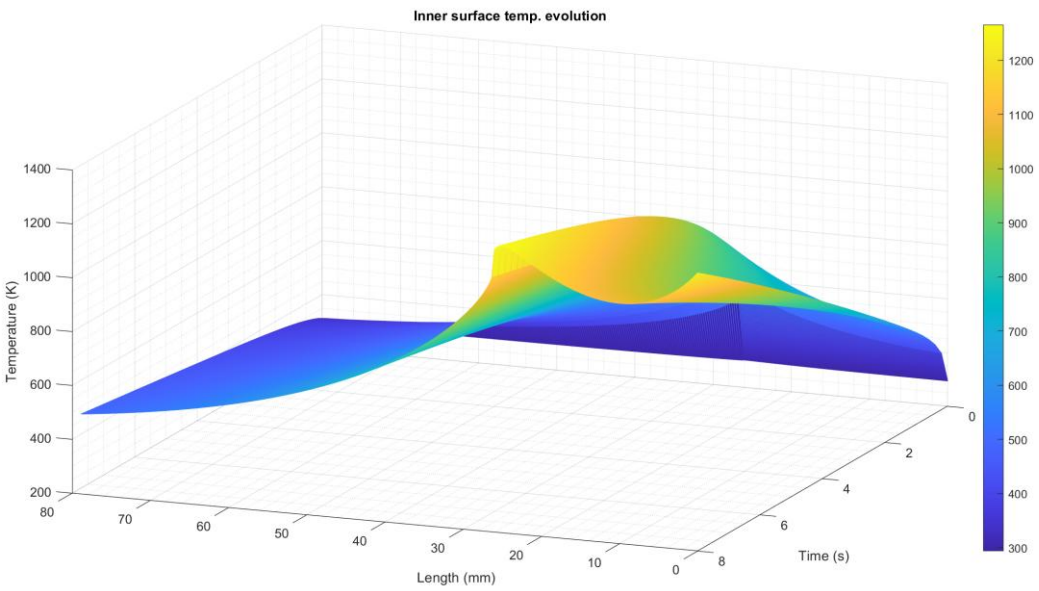

Figure 6.1: Inner surface temperature evolution (1st 3D View).

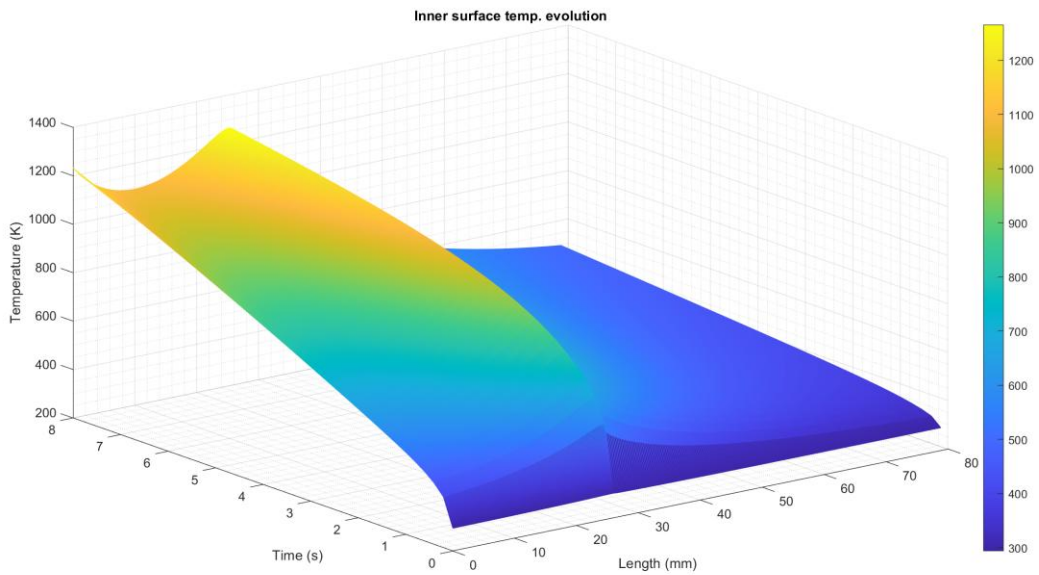

Figure 6.2: Inner surface temperature evolution (2nd 3D view).

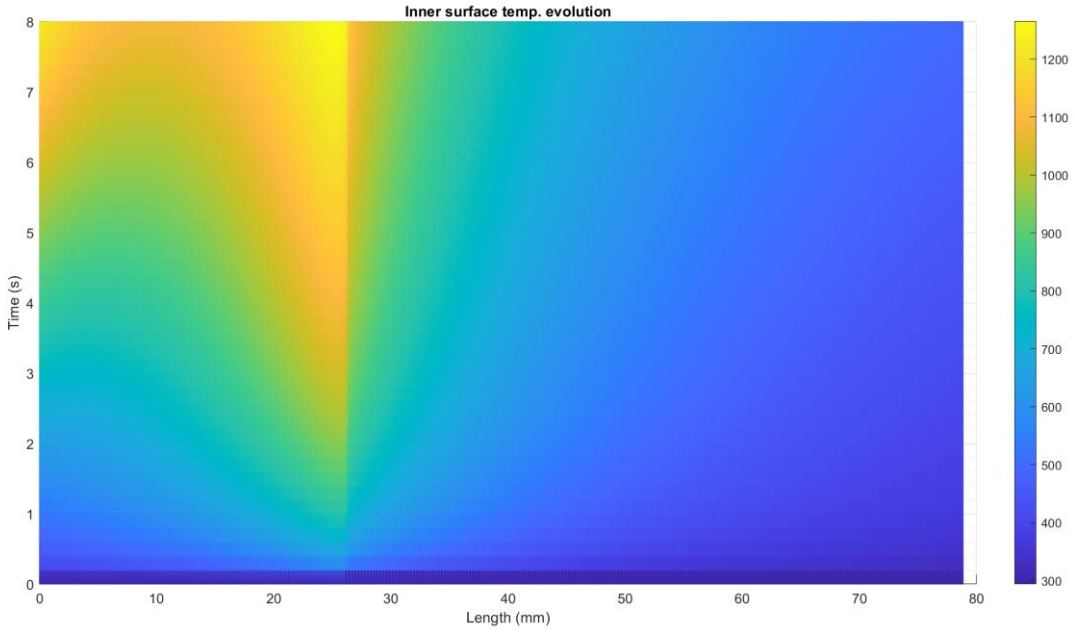

Figure 6.3: Inner surface temperature evolution (Top 2D view).

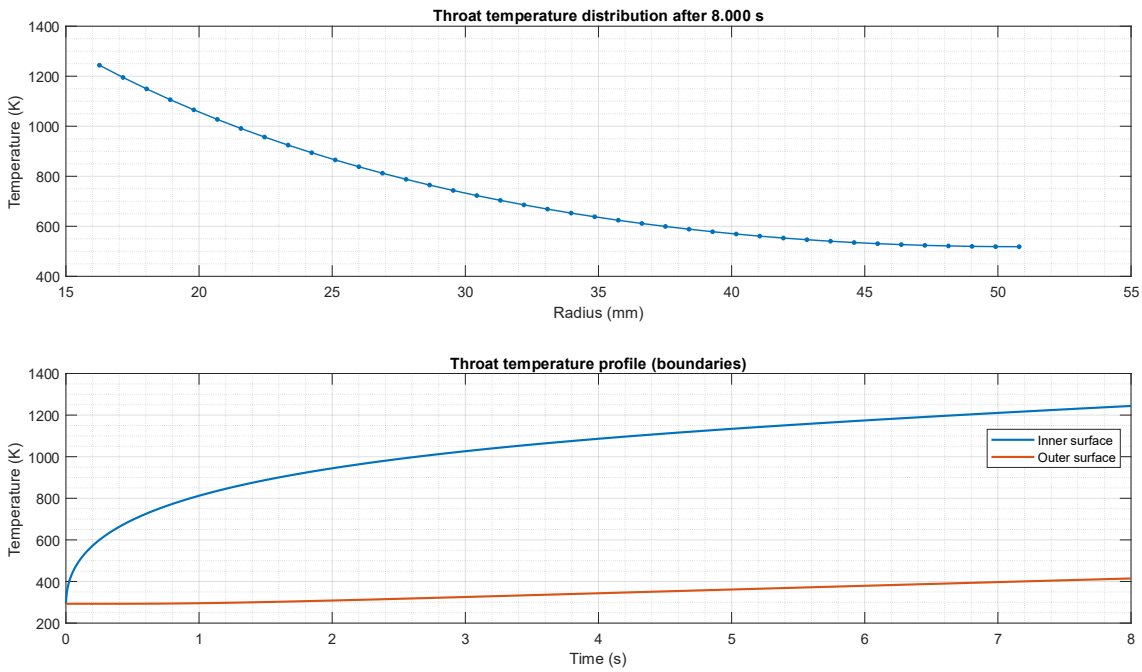

Figure 6.4: Throat temperature.

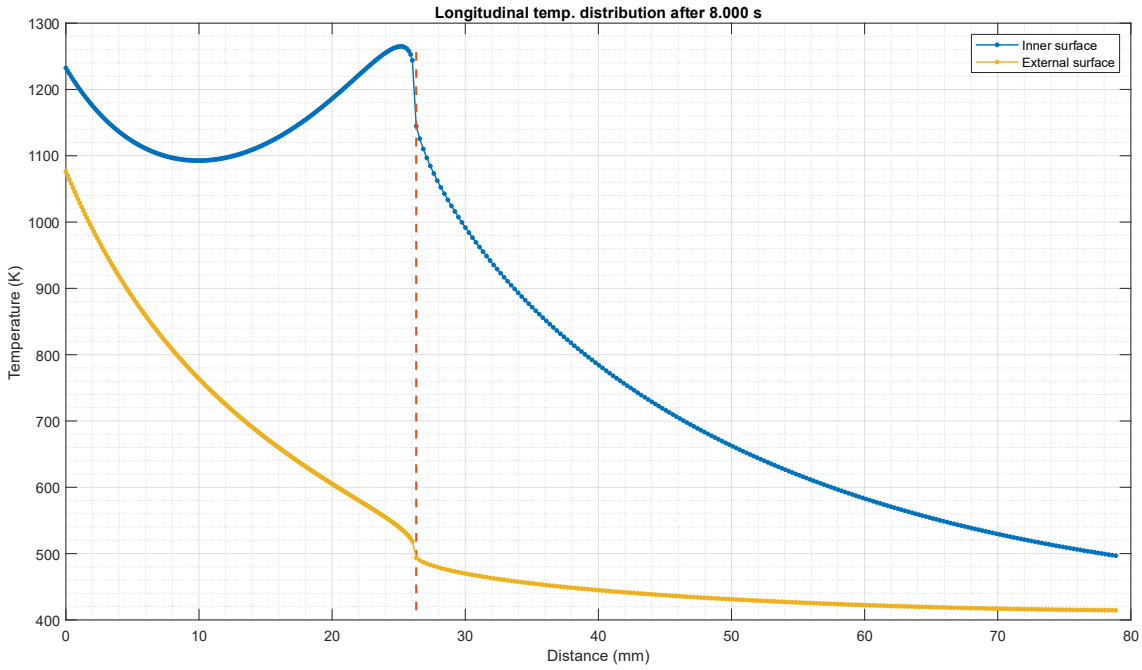

Figure 6.5: Longitudinal temperature distribution.

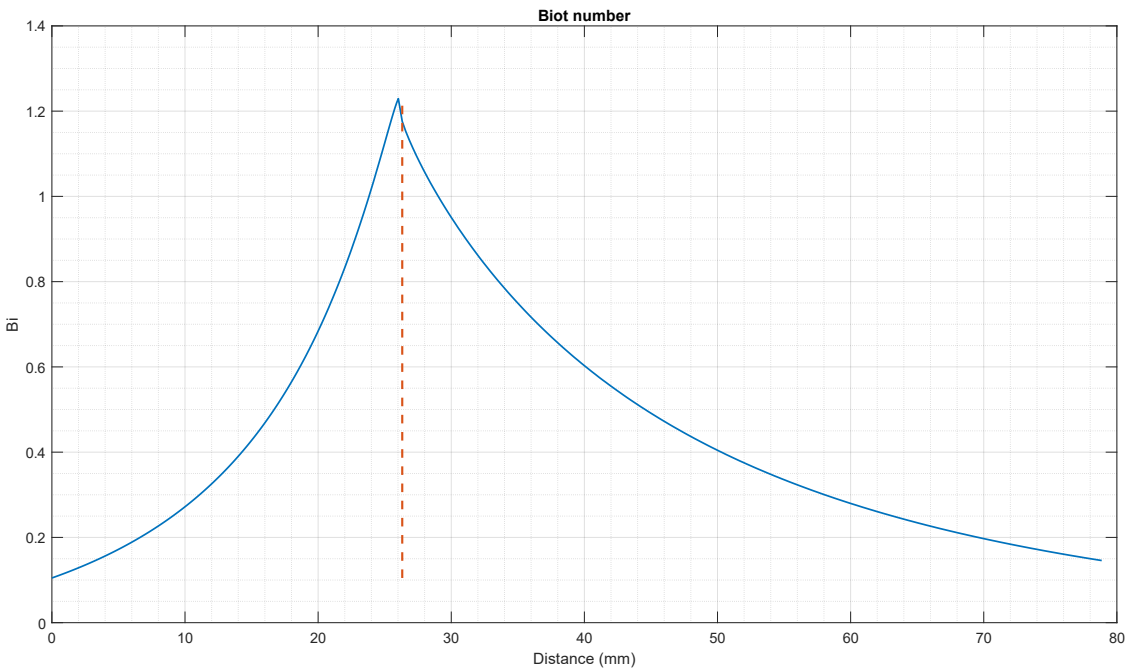

Figure 6.6: Biot number.

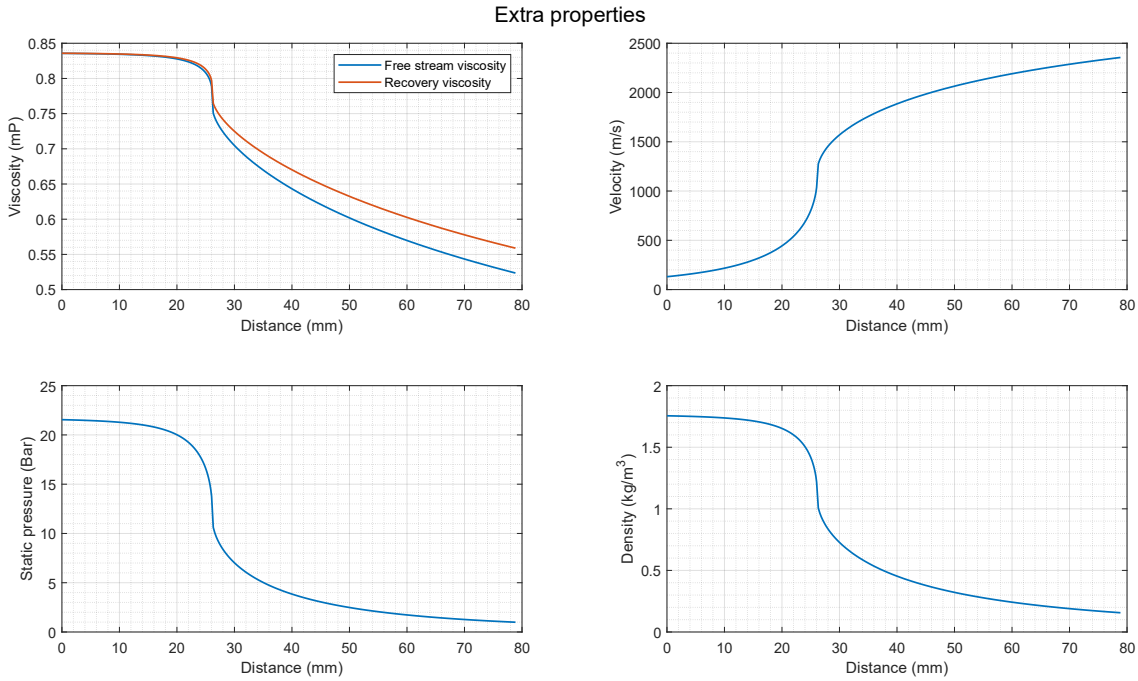

Figure 6.7: Nozzle extra properties.